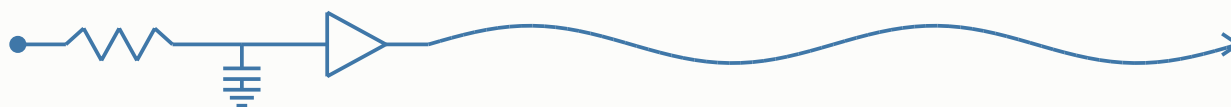


THE SUMMER EE GLOW-UP

VOL. 2 | 10 MORE RESOURCES

10 more resources and exactly how to use each one.
If you ran the first list, this is where you go deeper.

```
buildlist $ start glow-up --vol 2 --resources 10  
loading deeper cuts ... ready.
```



READ THIS FIRST

This is volume two. Ten more of the best, with zero overlap from the first list.

The first guide got you building. This one goes deeper, into the analog, the signals, the math, and the bench skill that separate a student from an engineer. Same format. Every resource has how to use it, the job it maps to, and a link. Tap any name to open it.

THE ONE RULE, STILL

Same rule as the first guide. Do not try to do all 10. Pick 3 that point at the same job and go deep. Depth is what gets you hired and what makes content nobody can fake.

THE 10 RESOURCES

Each one with what it is, how to actually use it, and the job it maps to. Links are clickable.

01 Practical Electronics for Inventors

BOOK | CORE | Scherz and Monk

> open [search for the book](#)

The desk reference that explains the whole field in plain language, from parts to working circuits.

HOW TO USE IT

- > Keep it open on the bench. When a part confuses you, read its section before you wire it.
- > Build the example circuits. Reading about a 555 timer is not the same as making one blink.
- > Use the chapter problems to find the gaps you did not know you had.

-> **The job this maps to:** Generalist hardware engineer. The book that makes you fluent across the whole stack.

02 Neso Academy

YOUTUBE | CORE | digital, signals, networks

> open youtube.com/@nesoacademy

Clear, structured lectures on the core EE topics your courses tend to rush through.

HOW TO USE IT

- > Fix weak spots fast. Pick the exact playlist for the topic that is hurting you.
- > Pause and redo every solved example by hand before moving on.
- > Treat the digital electronics and network theory playlists as your backup textbook.

-> **The job this maps to:** Any EE role. The fastest way to patch the fundamentals you are missing.

03 Razavi Electronics

YOUTUBE | ADVANCED | Behzad Razavi, UCLA

> open [Razavi lectures on YouTube](#)

The full UCLA electronics course from one of the people who literally wrote the analog textbooks.

HOW TO USE IT

- > Start at lecture one and go in order. This is a real course, not a clip to skim.
- > Work the board math along with him. Pause, derive, then continue.
- > This is your jump from hobby analog to real device-level understanding.

-> **The job this maps to:** Analog and IC design. The deep end most students never get near.

04 LTspice

TOOL | CORE | analog simulation

> open [analog.com LTspice](#)

The free SPICE simulator that working analog and power engineers actually use.

HOW TO USE IT

- > Simulate before you solder. Build the circuit, run a transient and an AC sweep, read the plots.
- > Recreate a circuit from Razavi or the book and confirm the theory on screen.
- > Learn the step and meas commands. Sweeping values is where the intuition clicks.

-> **The job this maps to:** Analog and power engineer. Simulation is assumed on the job, not optional.

05 Wokwi

TOOL | INTERMEDIATE | embedded simulation

> open wokwi.com

A browser simulator for Arduino, ESP32, and Pi Pico with real parts and no hardware needed.

HOW TO USE IT

- > Prototype firmware with zero setup. Wire a sensor and an MCU in the browser and run the code.
- > Test logic before your parts arrive, then move the same code to real hardware.
- > Build something small and shareable, then drop the link in a post.

-> **The job this maps to:** Embedded and IoT developer. Proves you can go from idea to running code fast.

06 Adafruit Learn

GUIDES | CORE | learn.adafruit.com

> open learn.adafruit.com

Thousands of free, tested project guides with wiring diagrams and full code.

HOW TO USE IT

- > Pick a project close to what you want to build and follow it end to end.
- > Then change the goal. Swap the sensor or the output so you are not just copying.
- > Steal their wiring and code patterns. They are clean and they work.

-> **The job this maps to:** Hardware and firmware engineer. The fastest on-ramp from parts to a finished build.

07 The Scientist and Engineer's Guide to DSP

BOOK | ADVANCED | free at dspguide.com

> open dspguide.com

The DSP book that teaches the concepts first and the heavy math second. Free, in full.

HOW TO USE IT

- > Read the early chapters on sampling and the Fourier transform until they feel obvious.
- > Run the example code on real data. DSP only makes sense when you see it work.
- > Come back to the filter chapters when you actually need to clean up a signal.

-> **The job this maps to:** DSP and signal processing engineer. The foundation under audio, RF, and sensors.

08 3Blue1Brown

YOUTUBE | CORE | the math intuition

> open youtube.com/@3blue1brown

The clearest visual explanations of the math EE runs on, especially Fourier and linear algebra.

HOW TO USE IT

- > Watch the Fourier and linear algebra series before your signals class, not after.
- > Do not just watch. Pause and predict what the animation will show next.
- > Use it to turn formulas you memorized into pictures you understand.

-> **The job this maps to:** Anything math-heavy. This separates people who use the math from people who fear it.



09 W2AEW

YOUTUBE | INTERMEDIATE | Alan Wolke, scope and bench

> open [w2aew on YouTube](#)

A Tektronix engineer showing how to actually use a scope and read real circuits on the bench.

HOW TO USE IT

- > Watch the oscilloscope basics videos before you touch the knobs on a real one.
- > Copy his measurement setups exactly, then try them on your own circuit.
- > This is how you stop guessing and start measuring like an engineer.

-> **The job this maps to:** Test and hardware engineer. Bench skill is what makes your designs trustworthy.



10 Hackaday and Hackaday.io

COMMUNITY | CORE | projects and inspiration

> open [hackaday.com](#) | [hackaday.io](#)

The daily feed of what people are building, plus a place to host and show your own projects.

HOW TO USE IT

- > Read it daily for ideas and to see what is actually possible right now.
- > Post your summer build on Hackaday.io with photos and notes. A public log is proof.
- > When you are stuck, search it. Someone has probably done a version of your project.

-> **The job this maps to:** Not a job, a multiplier. Building in public is how people find you and your work.

THE 12-WEEK STACK

How to combine them so the summer actually compounds. Each phase ends with something you ship.

PHASE	DO THIS	YOU SHIP
FOUNDATION Weeks 1-4	Practical Electronics for Inventors as your reference, Neso Academy for weak spots. LTspice open daily.	A circuit you simulated, breadboarded, and can explain.
DEPTH Weeks 5-8	Razavi for analog. 3Blue1Brown and The DSP Guide for signals. Verify on a scope with W2AEW's methods.	One circuit you designed, simulated, built, and measured.
BUILD Weeks 9-12	Prototype fast in Wokwi and Adafruit Learn, then post the project on Hackaday.	A working project and a public build log.

PICK YOUR LANE

Same move as last time. Three resources, one direction. Hackaday rides along no matter which you pick.

THE ANALOG BENCH LANE

You want to design analog and prove it on a scope.

- > Practical Electronics for Inventors
- > Razavi Electronics
- > W2AEW

Outcome: you design an analog circuit and verify it like an engineer.

THE SIM AND SIGNALS LANE

You want the math and the simulation to click.

- > LTspice
- > The DSP Guide
- > 3Blue1Brown

Outcome: you simulate circuits and understand the math behind every signal.

THE BUILD FAST LANE

You want to prototype and ship quickly.

- > Wokwi
- > Adafruit Learn
- > Neso Academy

Outcome: you go from idea to a working, shareable build fast.

THE GLOW-UP IS NOT THIS PDF

Two guides in, you have no shortage of resources. The shortage is reps. Pick your lane today and start phase one this week. The boards on your bench in September are the only proof that counts.

KEEP THE MOMENTUM

> THE BUILD LIST

The newsletter. Weekly build breakdowns and the projects you should not be able to make at home.

> FOLLOW THE BUILD

@1nska on Instagram. Daily build logs and the projects behind every guide.

Build in public. Ship by September.

Follow @1nska if you're building the future, not just studying for it.